



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PRE-SCORE
08-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

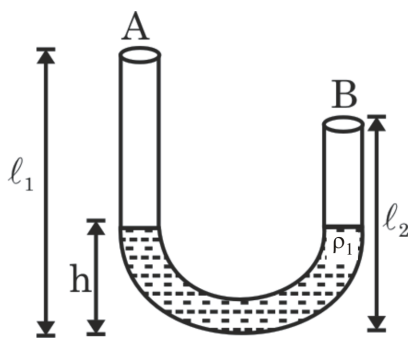
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.1 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 1%, the relative percentage error in the density is close to:
 (A) 0.9% (B) 2.4% (C) 3.1% (D) 4.2%
2. A tube U-shaped tube has a uniform cross-section with arm length l_1 and l_2 ($l_1 > l_2$). Tube has a liquid of density ρ_1 filled to a height h . Another liquid of density $\rho_2 = \frac{\rho_1}{4}$ is poured in arm A. Both liquids are immiscible. The length of the second liquid that should be poured in A so that first overtone of A is in unison with fundamental tone of B is



- (A) $2(l_1 - 3l_2 - 2h)$ (B) $2(l_1 - 3l_2 + 2h)$
 (C) $2(l_1 + 3l_2 + 2h)$ (D) None of these

Space for Rough Work

3. A gas is separated by a thin plane membrane, where temperature at one side is 7°C and that at other side is 47°C . If a narrow beam of sound is incident at an angle of 30° from cooler side, then the angle of refraction at hotter side is :-

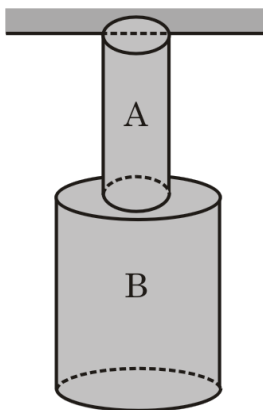
(A) $\sin^{-1} \sqrt{\frac{7}{8}}$

(B) $\sin^{-1} \sqrt{\frac{5}{7}}$

(C) $\sin^{-1} \sqrt{\frac{3}{7}}$

(D) $\sin^{-1} \sqrt{\frac{2}{7}}$

4. Two cylinders of same material and of same length are joined end to end as shown in figure. The upper end of A is rigidly fixed. The radii of A and B are in ratio 1 : 2. If the lower end of B is twisted by an angle θ , the angle of twist of cylinder A, is



(A) $\frac{15}{16} \theta$

(B) $\frac{16}{15} \theta$

(C) $\frac{17}{16} \theta$

(D) $\frac{16}{17} \theta$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

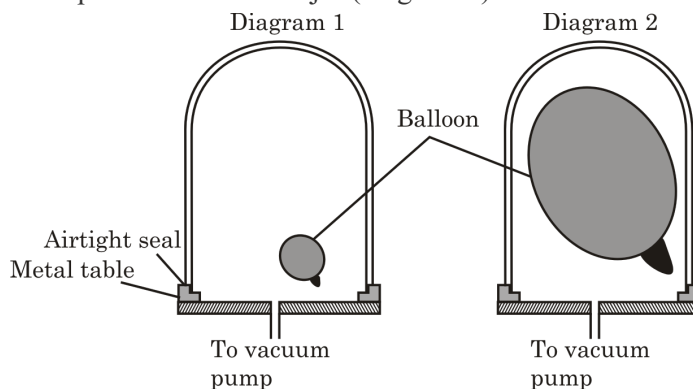
Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Ice at -20°C is filled upto height $h = 10\text{ cm}$ in a cylindrical vessel of uniform cross-section. Another identical vessel contains water at 0°C upto the same height. Now water of the second vessel is poured into the first vessel and it is found that level of upper surface falls through 0.5 cm when the thermal equilibrium is achieved. Neglect heat loss, variation of density temperature and thermal capacities of the vessel. Take $\rho_{\text{water}} = 1\text{ g/cc}$, $\rho_{\text{ice}} = 0.9\text{ g/cc}$, $S_{\text{water}} = 1\text{ cal/g}^{\circ}\text{C}$, $L_f = 80\text{ cal/g}$. For this situation mark out the correct statement(s).
- (A) Value of θ is 45°C
- (B) At thermal equilibrium, the temperature is 0°C .
- (C) In equilibrium the vessel contains a mixture of ice and water.
- (D) Value of θ is 86°C

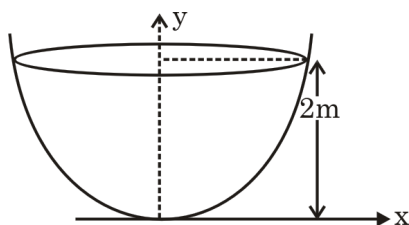
Space for Rough Work

6. A partially inflated balloon is placed under a bell jar (diagram 1).



A vacuum pump is turned on for several minutes and the volume of the balloon increases (diagram 2).

- (A) Pressure inside the bell jar increases. (B) Pressure inside the bell jar decreases.
(C) Pressure inside the balloon increases. (D) Pressure inside the balloon decreases.
7. A paraboloid shape container is placed at origin as shown in figure, filled with liquid up to a height 2 m. Equation of parabola is given as $x^2 = 2y$. A small opening is made at origin of area 10 cm^2 , then ($\pi = \sqrt{10}$).



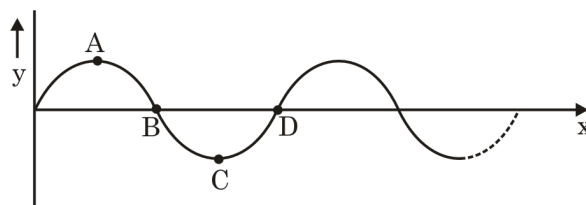
- (A) Speed of efflux when height of liquid is 0.8 m is 2 m/s.
(B) Speed of efflux when height of liquid is 0.2 m is 2 m/s.
(C) Time to empty the container is $\frac{20}{27}$ hr
(D) Time to empty the container is $\frac{27}{20}$ hr.

Space for Rough Work

8. Consider two media A and B with refractive indices μ_1 and μ_2 respectively. A ray of light travels from A to B. Its directions in the two media are given by the unit vectors, $\vec{r}_A = a\hat{i} + b\hat{j}$ and $\vec{r}_B = \alpha\hat{i} + \beta\hat{j}$ respectively where $\hat{i}$ & $\hat{j}$ are unit vectors in the x and y directions. Choose the correct statement(s) :-
- (A) If the media is separated by the x-z plane then $\mu_1 a = \mu_2 \alpha$
- (B) If the media is separated by the y-z plane then $\mu_1 a = \mu_2 \alpha$
- (C) If the media is separated by the x-z plane then $\mu_1 b = \mu_2 \beta$
- (D) If the media is separated by the y-z plane then $\mu_1 b = \mu_2 \beta$
9. Consider a long narrow cylinder of cross-section A filled with a compressible liquid upto a height h whose density ρ is a function of pressure P(z) as $\rho(z) = \frac{\rho_0}{2} \left(1 + \frac{P(z)}{P_0} \right)$, where P_0 and ρ_0 are constants. The depth z is measured from the free surface of the liquid where pressure is equal to atmospheric pressure (P_{atm}). Choose the correct options.
- (A) Density varies with depth as $\rho(z) = \frac{\rho_0 (P_0 + P_{\text{atm}})}{2P_0} e^{\rho_0 g z / 2P_0}$
- (B) Mass of the liquid in container is $\frac{A}{g} (P_0 + P_{\text{atm}}) (e^{\rho_0 g h / 2P_0} - 1)$
- (C) Pressure varies with depth as $P(z) = P_{\text{atm}} e^{\rho_0 g z / 2P_0} - P_0$
- (D) $\frac{dP(z)}{dz} = \frac{\rho_0}{2} \left(1 + \frac{P(z)}{P_0} \right) g$

Space for Rough Work

10. Figure shows a wave motion in which displacements of particles at an instant is represented as shown. Choose the correct options.



- (A) If the wave represents a transverse wave in a stretched string, then an element in region B has max. kinetic energy.
- (B) If it represents a transverse wave in a stretched string, then an element in the region B has max potential energy.
- (C) If it represents sound wave travelling in air, point D represents rarefaction.
- (D) If it represents sound wave travelling in air, point A has normal atmospheric pressure.

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

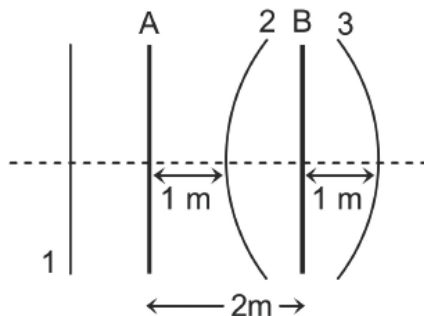
- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

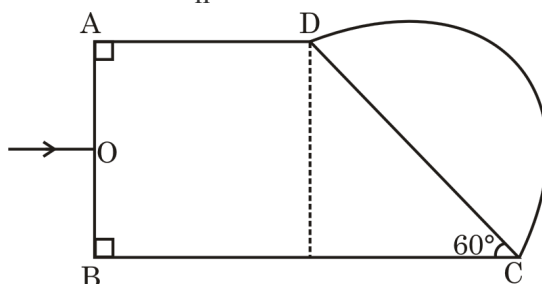
1. Each of the two strings of length 51.6 cm and 49.1 cm are tensioned separately by 20N force. They are fixed at both ends. Mass per unit length of both the strings is same and equal to 1 g/m. When both the strings vibrate simultaneously in fundamental mode the number of beats is :
2. A and B are thin lenses. 1, 2 and 3 are representative wave fronts of incident and refracted monochromatic light with distances from the lenses as shown. The radii of curvature of wave fronts 2 and 3 are 4m and 7m respectively. Find the focal length (modulus value) of lens B in meter.



3. The shear stress in a liquid flowing through a pipe of radius R depends on radial distance from centre given by $\tau = -2r$ and volume of liquid flowing through pipe per second $\frac{k\pi R^4}{8\eta}$. Find the value of k. (η -coefficient of viscosity).

Space for Rough Work

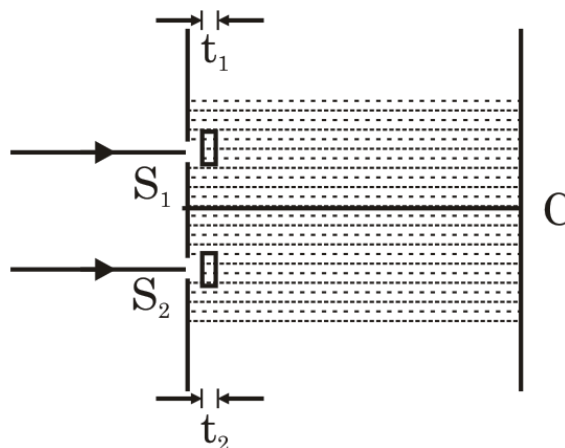
4. A slab and hemisphere are joined together as shown in figure. The ratio of refractive indices of the slab and hemisphere is $\sqrt{3} : 1$. A ray is normally incident at 'O' on the vertical face AB of the block such that $AO = OB$. The ray will suffer total deviation $\frac{60}{n}$ degrees after all refractions. Find value of n .



5. A radioactive element X undergoes β -decay with a half-life of T days to form a stable element Y. Initially, a vessel contains mass m of 100% pure X. After every time interval of exactly T days, half of the sample in the vessel (by mass) is removed. The mass of the element Y in the vessel after $4T$ days from the start (just before half the sample in the vessel is removed) is m_f . The value of $\frac{m_f}{4m} \times 100$ (to the nearest integer) is _____.
6. Find the quantum number n corresponding to n^{th} excited state of He^+ ion if on transition to the ground state the ion emits two photons in succession with wavelengths 108.5 nm and 30.4 nm. The ionization energy of the hydrogen atom is 13.6 eV.
7. One mole of Argon gas is enclosed in a container at initial pressure P_0 and volume V_0 . The gas is taken to final pressure $\frac{P_0}{2}$ and volume $3V_0$ through two processes in succession : an isobaric expansion, followed by an isothermal expansion. If the work done by the gas in the two processes is W_1 and W_2 respectively, and the ratio $\frac{W_2}{W_1} = \beta \ln(2)$, then β is _____

Space for Rough Work

8. A screen is at a distance $D = 80$ cm from a diaphragm having two narrow slits S_1 and S_2 which are $d = 2$ mm apart. Slit S_1 is covered by a transparent sheet of thickness $t_1 = 2.5$ μm and S_2 by another sheet of thickness $t_2 = 1.25$ μm as shown in figure. Both sheets are made of same material having refractive index $\mu = 1.40$ and water is filled in space between diaphragm and screen. A monochromatic light beam of wavelength $\lambda = 5000$ $\text{\AA}$ is incident normally on the diaphragm. Assuming intensity of beam to be uniform, calculate ratio of intensity at central point on screen to individual beams through slits. ($\mu_w = 4/3$).



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. In a solution containing 0.5M - CH_3COONa , 0.2 mole HNO_3 is added in 1.0 L of solution. The pH of resulting solution is [K_b of $\text{CH}_3\text{COO}^- = 5 \times 10^{-10}$, $\log 2 = 0.30$, $\log 3 = 0.48$]
(A) 4.70 (B) 4.88 (C) 9.48 (D) 4.52
2. The work function of a metal is 4.0 eV. Electromagnetic radiation of wavelength 206.67 nm fall on the surface of this metal. The minimum de-Broglie wavelength of photoelectron ejected is :
(A) 8.66 Å (B) 5.0 Å
(C) 6.12 Å (D) 7.45 Å
3. Calculate the maximum pH of solution at which 1.0 mole of $\text{Al}(\text{OH})_3(\text{s})$ dissolves completely in 2.0 L water. [$\log 2 = 0.30$]
 $\text{Al}(\text{OH})_3(\text{s}) \rightleftharpoons \text{Al}^{3+}(\text{aq}) + 3\text{OH}^-(\text{aq}); K = 4 \times 10^{-18}$
(A) 5.70 (B) 8.30
(C) 8.20 (D) 9.30
4. The minimum relative humidity to make $\text{A.H}_2\text{O}(\text{s})$ hygroscopic at 25°C , is
 $\text{A.3H}_2\text{O}(\text{s}) \rightleftharpoons \text{A.H}_2\text{O}(\text{s}) + 2\text{H}_2\text{O}(\text{g}); K_p = 9 \times 10^{-4} \text{ atm}^2$
At 25°C , the vapour pressure of water is 0.04 atm.
(A) 25% (B) 50% (C) 75% (D) 10%

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

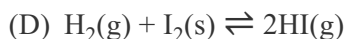
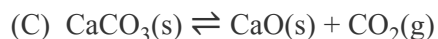
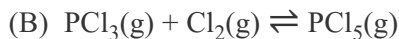
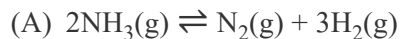
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. For which of the following reaction(s), $K_p < K_c$ at 25°C ?



6. For the reaction : $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$, $K_p = 80$ atm. If in a sealed rigid vessel, N_2O_4 and NO_2 gases are taken at 0.2 atm and 2.0 atm, respectively and maintained at constant temperature, the correct information(s) about the changes in system with time is/are :

(A) Colour of system deepens

(B) $P_{\text{N}_2\text{O}_4}$ increases

(C) P_{NO_2} increases

(D) Density of system increases

Space for Rough Work

7. The correct information(s) about $4d_{z^2}$ orbital of H-atom is/are :
- (A) It does not have radial node. (B) It has two angular nodes.
(C) It does not have planar nodal surface. (D) It has one spherical nodal surface.
8. The correct information(s) for 100 ml of 0.5 M NH_4Cl solution is/are
(K_b of $\text{NH}_4\text{OH} = 1.8 \times 10^{-5}$, $\log 2 = 0.30$, $\log 3 = 0.48$).
- (A) Degree of hydrolysis = $\frac{10^{-4}}{3}$ (B) pH = 4.78
(C) pOH = 4.78 (D) Moles of $\text{NH}_4\text{OH} = \frac{10^{-5}}{6}$
9. A solution contains 0.5 M - Ag^+ and 0.2 M - Hg_2^{2+} . The ions may be precipitated selectively, adding Cl^- ion in the solution, starting with a very small amount. Select the correct information(s).
[K_{sp} of $\text{AgCl} = 2 \times 10^{-10}$, K_{sp} of $\text{Hg}_2\text{Cl}_2 = 8 \times 10^{-17}$]
- (A) Hg_2^{2+} ion will precipitate first.
(B) Ag^+ ion will precipitate first.
(C) 98% of Ag^+ ion will precipitate before the start of precipitation of Hg_2^{2+} ion.
(D) Almost 100% of Hg_2^{2+} ion will precipitate before the start of precipitation of Ag^+ ion.
10. Solids A and B dissociates as :
 $\text{A(s)} \rightleftharpoons 2\text{C(g)} + \text{D(g)}; K_{P_1}$
 $\text{B(s)} \rightleftharpoons 2\text{C(g)} + \text{E(g)}; K_{P_2}$
The total equilibrium pressure over excess solids A and B are 60 torr and 90 torr, respectively. The incorrect information(s) is/are :
- (A) $K_{P_1} = 400 \text{ torr}^3$
(B) $K_{P_2} = 2.7 \times 10^4 \text{ torr}^3$
(C) If solids A and B are taken in the same vessel, then the molar ratio of D(g) and E(g) is 8 : 27.
(D) If solids A and B are taken in the same vessel, then the total equilibrium pressure of gases becomes 120 torr.

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. pH of 0.05 M - HA solution is (K_a of HA = 2×10^{-5}).
2. The radius of 4th orbit for a H-like atom (atomic number = Z) is 2.82 Å. The value of 'Z' is.
3. For the reaction : $A(aq) \rightleftharpoons 2B(aq)$; $K_C = 0.4$ M. If the equilibrium concentration of B(aq) is 0.2 M, the moles of 'A' dissolved initially in 10 L water is.
4. The solubility product of $A_2B_3(s)$ is $1.08 \times 10^{-18} \text{ M}^5$. the millimoles of $A_2B_3(s)$ dissolved in 10 L water is.
5. The solubility of $Cd(OH)_2(s)$ in a basic solution is due to its dissociation as well as complex formation :
 $Cd(OH)_2(s) \rightleftharpoons Cd^{2+}(aq) + 2OH^-(aq)$; $K_1 = 10^{-14} \text{ M}^3$
 $Cd(OH)_2(s) + 2OH^-(aq) \rightleftharpoons Cd(OH)_4^{2-}(aq)$; $K_2 = 10^2 \text{ M}^{-1}$
 The pOH of solution at which the solubility of $Cd(OH)_2(s)$ will be minimum is.
6. A sample of hydrogen atoms have all the atoms in a particular excited state of $n = 5$. If all the atoms de-excite to the ground state but no radiation is emitted out in Balmer Series, the total number of spectral lines emitted out will be (only the lines with different wavelengths are counted as spectral lines).
7. Find the principal quantum number for the electron in He^+ ion with de-Broglie wavelength equal to 8.3 Å.
8. The pH range of a basic indicator, BOH, is 7.0 to 9.0. the value of pK_b of BOH is.

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. $(x-1)(y-2) = 5$ and $(x-1)^2 + (y+2)^2 = r^2$ intersect at four points A, B, C, D and if centroid of ΔABC lies on line $y = 3x - 4$, then locus of D is
- (A) $y = 3x$ (B) $x^2 + y^2 + 3x + 1 = 0$
(C) $3y = x + 1$ (D) $y = 3x + 1$
2. Two circles with radii $\sqrt{2}$ cm and 1 cm meet at a point A, whose centres lie on x-axis. The distance between their centres is 2 cm. The chord AC of the larger circle cuts the smaller circle at a point B and is bisected by that point. The length of chord AC is
- (A) $\frac{\sqrt{5}}{2}$ cm (B) $\sqrt{\frac{7}{2}}$ cm (C) $\frac{\sqrt{3}}{2}$ cm (D) $\sqrt{\frac{5}{2}}$ cm
3. $e^{|\sin x|} + e^{-|\sin x|} + 4a = 0$ will have exactly four different solutions in $[0, 2\pi]$ if
- (A) $a \in \left[-\frac{e}{4}, -\frac{1}{4}\right]$ (B) $a \in \mathbb{R}$
(C) $a \in \left[\frac{-1 - e^2}{4e}, \infty\right]$ (D) None of these
4. The value of b, such that the equation $\frac{b \cos x}{2 \cos 2x - 1} = \frac{b + \sin x}{(\cos^2 x - 3 \sin^2 x) \tan x}$ posses solutions, belong to the set
- (A) $\left(-\infty, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \infty\right)$ (C) $(-\infty, \infty)$ (D) $\left(-\infty, \frac{1}{2}\right) \cup (1, \infty)$
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A straight line passing through the point A(-2,-3) cuts lines ' $x + 3y = 9$ ' and $x + y + 1 = 0$ at B and C respectively. If AB. AC = 20, then equation of possible line is

(A) $x - y = 1$

(B) $x - y + 1 = 0$

(C) $3x - y + 3 = 0$

(D) $3x - y = 3$

Space for Rough Work

6. The equation of the bisectors of the angles between the two intersecting lines $\frac{x-3}{\cos \theta} = \frac{y+5}{\sin \theta}$ and $\frac{x-3}{\cos \phi} = \frac{y+5}{\sin \phi}$ are $\frac{x-3}{\cos \alpha} = \frac{y+5}{\sin \alpha}$ and $\frac{x-3}{\beta} = \frac{y+5}{\gamma}$, then
- (A) $\alpha = \frac{\theta + \phi}{2}$ (B) $\beta = -\sin \alpha$
 (C) $\gamma = \cos \alpha$ (D) $\beta = \sin \alpha$
7. For $\alpha \in \mathbb{N}$, consider a relation R on $\mathbb{N}$ given by $R = \{(x, y) : 3x + \alpha y \text{ is a multiple of } 7\}$. The relation R is an equivalence relation if and only if :
- (A) $\alpha = 14$
 (B) α is a multiple of 4
 (C) 4 is the remainder when α is divided by 10
 (D) 4 is the remainder when α is divided by 7
8. Let $S_1 = x^2 + y^2 - 6 = 0$, $S_2 = x^2 + y^2 - 6x + 8 = 0$, $S_3 = x^2 + y^2 + 2x + 2ky + 6 = 0$ and $S_4 = x^2 + y^2 + 2ky + k = 0$, then which of the following is/are true -
- (A) If S_3 and S_4 intersect orthogonally, then value of k may be 2.
 (B) If S_3 and S_4 intersect orthogonally, then value of k may be $-\frac{3}{2}$.
 (C) The equation of circle passing through (1,1) and intersection of S_1 & S_2 is $x^2 + y^2 - 3x + 1 = 0$
 (D) The equation of circle passing through (1,1) and intersection of S_1 & S_2 is $x^2 + y^2 - 6x + 4 = 0$

Space for Rough Work

9. Which of the following is/are correct ?

(A) If the image of (3, 8) in the line $x + 3y = 7$ is (a,b), then $a - b$ is equal to 3

(B) If the points (ℓ, ℓ) , $(\ell, 2\ell)$ & $(2\ell, \ell)$ enclose a triangle of area 18 sq. units and (a,b) is its centroid then $\frac{|a| + |b|}{4}$ is equal to 4

(C) If the area enclosed between the pair of tangents drawn from (4,5) to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ & the given circle is equal to $k - 9t$ where $\tan t = \frac{4}{3}$ then $\frac{k}{2}$ is equal to 6

(D) Area of the triangle formed by the lines $x^2y + x^2 - y^3 + 3y^2 - 4 = 0$ is equal to 9

10. Consider the curve

$$S = (\sin\theta)x^2 + 2(\cos\theta)xy + (\sin\theta)y^2 + 2(\cos\theta)x + 2(\cos\theta)y + \sin\theta = 0$$

Now, choose the correct statement(s)

(A) If $\theta = \frac{3\pi}{4}$, then $S = 0$ represents a pair of straight lines.

(B) If $S = 0$ represents a pair of real straight lines, then $\theta = n\pi + \frac{\pi}{4}$, $n \in \mathbb{Z}$

(C) If $S = 0$ represents a pair of real straight lines, then they are parallel.

(D) There exist some θ for which $S = 0$ represents no real locus.

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

-
1. A straight line passing through origin O meets the lines $4x + 2y = 3$ and $2x + y + 6 = 0$ at the points P and Q respectively, then the point O divides the segment PQ in the ratio 1 : k then k is equal to _____
 2. Three circles lie on a plane so that each of them externally touches the other two. Two of them has radius 3, the third having radius unity. If A, B and C are the points of tangency of those circles and the area of the triangle ABC is $\frac{a\sqrt{b}}{c}$ then value of $c - a - b$ is, (where a, b and c are coprime)
 3. If $x^4 + 3 \cos(ax^2 + bx + c) = 2(x^2 - 2)$ has two solutions with $a, b, c \in (2, 5)$ then the maximum value of $\frac{ac}{b^2}$ is _____
-

Space for Rough Work

4. Find the number of common solution(s) of the trigonometric equations $\cos 2x + (1 - \sqrt{3}) = (2 - \sqrt{3})\cos x$ and $\sin 3x = 2 \sin x$ which satisfy the inequality $\sqrt{3}\tan x - 1 \geq 0$ in $[0, 5\pi]$.
5. Number of value of x in $[0, 2\pi]$ for which $2^{(2\sin x - 1)(\sin x - 1)} + 2^{(2\sin x + 1)(2 - \sin x)} = 9$, is
6. Let C_1 and C_2 are two circles $x^2 + y^2 + 10x - 24y - 87 = 0$ and $x^2 + y^2 - 10x - 24y + 153 = 0$ respectively. Let m be the smallest positive value of a for which the line $y = ax$ contains the centre of a circle that is internally tangent to C_1 and externally tangent to C_2 , then the value of $[m]$ is (where $[.]$ denotes greatest integer function)
7. If $(\lambda, \lambda + 1)$ is an interior point of $\triangle ABC$, where $A \equiv (0, 3)$, $B \equiv (-2, 0)$ and $C \equiv (6, 1)$, then the number of integral value of λ is :
8. Let the mean and the variance of 20 observations $x_1, x_2, \dots, x_{20}$ be 15 and 9, respectively. For $\alpha \in \mathbb{R}$, if the mean of $(x_1 + \alpha)^2, (x_2 + \alpha)^2, \dots, (x_{20} + \alpha)^2$ is 178, then the square of the maximum value of α is equal to _____.

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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